

LESSON 6-6 • SESSION 1 • GUIDED NOTES

Simplifying to Show Equivalence

Learning target – I can simplify an expression by combining like terms to test whether two expressions are equivalent.
6.EE.A.3

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
like terms <i>términos semejantes</i>	Terms with the same _____ part.	$7c$ and $6c$
combine <i>combinar</i>	Add or subtract like terms into ____ term.	$7c + 6c = 13c$
equivalent expressions <i>expresiones equivalentes</i>	Expressions that have the same _____ for every number.	$7c + 9 + 6c$ and $13c + 9$

KEY IDEA*complete each sentence*

- Like terms have the same _____ part; $4x$ and 4 are not like terms.
- Combine like terms by adding their _____: $4x + 9x = 13x$.
- Two expressions are equivalent if they simplify to the _____ expression.

WORKED EXAMPLE*fill in the blanks as we solve*Simplify $9b + 4 + 3b + 1$.**Step 1** The like terms with b are $9b$ and ____.**Step 2** Combine them: $9b + 3b =$ ____.**Step 3** Combine the constants: $4 + 1 =$ ____.**Answer** $9b + 4 + 3b + 1 =$ _____**YOU TRY***use the example as a model***1.** Simplify.

a. $6a + 2a =$ _____

b. $x + 4x + 3 =$ _____

2. Are $2n + 7 + 3n$ and $5n + 7$ equivalent? Simplify to decide.*Show your work.***Equivalent?** _____**SUM IT UP***in your own words**I can combine $3x$ and $5x$ but not $3x$ and 5 because ____.*

LESSON 6-6 • SESSION 2 • GUIDED NOTES

Testing and Justifying Equivalence

Learning target – I can test whether two expressions are equivalent by substituting values and by reasoning. · 6.EE.A.4

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
equivalent expressions <i>expresiones equivalentes</i>	Expressions with the same value for ____ value of the variable.	$2(y + 7)$ and $2y + 14$
counterexample <i>contraejemplo</i>	One test value that shows a claim is ____.	$x = 1$: $2x + 8 = 10$, $4x = 4$
Distributive Property <i>propiedad distributiva</i>	Multiply the number outside by ____ term inside the parentheses.	$4(n + 3) = 4n + 12$
Commutative Property <i>propiedad conmutativa</i>	You can change the ____ when you add or multiply.	$a + b = b + a$

KEY IDEA*complete each sentence*

- To test, substitute the ____ value into both expressions.
- One value that gives different results proves they are ____ equivalent.
- Matching values do not prove it; ____ with properties.

WORKED EXAMPLE*fill in the blanks as we solve*Are $2x + 8$ and $4x$ equivalent?**Step 1** Try $x = 4$: $2 \times 4 + 8 = 16$ and $4 \times 4 = \underline{\hspace{1cm}}$. One match is not proof.**Step 2** Try $x = 1$: $2 \times 1 + 8 = \underline{\hspace{1cm}}$ and $4 \times 1 = 4$.**Step 3** $10 \neq 4$, so $x = 1$ is a ____.**Answer** They are ____ equivalent.**YOU TRY***use the example as a model***1.** Evaluate when $a = 3$.

a. $2(a + 1) = \underline{\hspace{2cm}}$

b. $2a + 1 = \underline{\hspace{2cm}}$

2. Are $5m$ and $m + 4$ equivalent? Test $m = 1$ and $m = 2$.*Show your work.***Equivalent?** _____**SUM IT UP***in your own words*

One matching test value does not prove two expressions are equivalent because ____.